

# Solution Requirements

|               |                                  |
|---------------|----------------------------------|
| Date          | 29 September 2026                |
| Team ID       | SWTID-2026-1393                  |
| Project Name  | EduGenie – AI Learning Assistant |
| Maximum Marks | 4 Marks                          |

**Define Functional and Non-Functional Requirements:**

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system. Non-Functional Requirements (NFR) specify criteria that judge how the system should operate.

**Step 1: Functional Requirements (FR)**

| S.No | Requirement Category              | Requirement Description                                                                                                                                                | Priority |
|------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1    | Authentication                    | Users can access the EduGenie learning assistant through the application interface; user input and session data should be handled securely.                            | High     |
| 2    | Authorization levels              | Student users can access learning features such as notes, quizzes, explanations and study support. Administrative access, if added, is restricted to authorized users. | Medium   |
| 3    | External interfaces               | Integrates with the AI model/API used by EduGenie to generate explanations, summaries, quizzes and learning assistance.                                                | High     |
| 4    | Transactions processing           | Accepts a learning request, sends the required prompt/data to the AI service, receives the generated result, and displays it to the user.                              | High     |
| 5    | Reporting                         | Provides generated notes, quiz results, explanations and learning-progress information in a readable format.                                                           | Medium   |
| 6    | Business rules, etc.              | The system should generate learning content from the user's topic or question and provide relevant explanations, quizzes and study support.                            | High     |
| 7    | Compliance to laws or regulations | Protect user data, avoid exposing API keys or sensitive configuration, and follow applicable data-privacy and responsible-AI requirements.                             | High     |
| 8    | Other                             | Provide an easy-to-use interface for learning assistance, including notes/summaries, quizzes, question answering and personalized learning support.                    | High     |

Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category               | Requirement Description                                                                                                   | Target Metric / Acceptance Criteria                                                                                         |
|------|----------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1    | Performance & Speed        | The interface should respond promptly and AI-generated results should be displayed without unnecessary delay.             | Normal page interactions should respond within 2 seconds; AI response time depends on the external AI service.              |
| 2    | Scalability                | The application should support growth in users, learning requests and stored learning data.                               | Design should allow additional users and requests without major redesign.                                                   |
| 3    | Security & Data Privacy    | Protect user inputs, credentials and configuration. Keep API keys and secrets out of source code and public repositories. | Secrets stored securely; no API key exposed in client-side code or repository.                                              |
| 4    | Reliability & Availability | The application should handle temporary service/API failures gracefully and provide a useful error message.               | Graceful error handling for failed AI requests; application remains usable for supported local features.                    |
| 5    | Usability & Accessibility  | The interface should be simple, readable and easy for students to navigate.                                               | A first-time student should be able to enter a topic/question and access the generated learning support without assistance. |
| 6    | Other                      | Maintainable and portable code structure with clear modules for learning features and future enhancements.                | Code organized into reusable modules; easy to update or extend with additional learning features.                           |

Team Members – Group 1

| No. | Member Name      | Group No. |
|-----|------------------|-----------|
| 1   | BALAMURUGAN M    | 1         |
| 2   | Aakash A         | 1         |
| 3   | Akash S          | 1         |
| 4   | Santhosh kumar R | 1         |
| 5   | Gunaseelan K     | 1         |